BrightLearn Data Analytics Exercise 6

Advanced SQL Date Functions

Table 1: Employees

emp-id	name	hire_date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

Q1: Add 6 months to each employee's hire date using DATEADD().

Expected columns: emp_id, name, hire_date, hire_plus_6_month

Code:

```
SELECT emp_id, name, hire_date
        DATEADD(MONTH, 6, hire_date) AS hire_plus_6_month
FROM Employees;
```

Output table:

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2: Students

student_id	name	dob
101	Maya	2005-08-10
102	Ethan	2004-12-01
103	Sienna	2006-03-15

Q2. Use DATEDIFF() to find age in days from dob to today.

Expected Columns: student_id, name, age_in_days

Code:

```
SELECT student_id, name,
        DATEDIFF(DAY, dob, CURRENT_DATE) AS age_in_days
FROM Students;
```

Output table

student_id	name	age_in_days
101	Maya	7540
102	Ethan	7800
103	Sienna	7300

Table 3: Events

event_id	event_name	event_date
1	Semina	2024-06-15
2	Workshop	2025-09-01
3	Hackathon	2025-01-20

Q3. Find how many days are left until each event using DATEDIFF().

Expected Columns: event_id, event_name, days_remaining

Code:

```
SELECT event_id, event_name
        DATEDIFF(DAY, CURRENT_DATE, event_date) AS
days_remaining
FROM events;
```

Output table.

event_id	event_name	days_remaining
1	Seminar	-306
2	Workshop	137
3	Hackathon	-87

Table 4: Invoices

invoice_id	issue_date	due_date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

Q4. Calculate the number of days between issue_date and due_date

Expected columns: invoce_id, issue_date, due_date, days_betu

Code:

```
SELECT invoice_id, issue_date, due_date,
        DATEDIFF(DAYS issue_date, due_date) AS days_betwee
FROM Invoinces;
```

Output table

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5: Courses

course_id	name	start_date
201	SQL Basics	2025-05-01 ~~start_date~~
202	Python	2025-06-01

Q5. Format start_date as 'Month YYYY' using TO_CHARC.
Expected Columns: course_id, name, formatted_date

Code:

```
SELECT course_id, name,
        TO_CHAR (start_date, 'Month YYYY') AS Formatt
_date
FROM Courses;
```

table:

course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

Table 6: Memberships

member_id	start_year	start_month	start_day
1	2023	5	10
2	2022	11	25

Q6. Create full date from parts using DATE_FROM_PARTS().
Expected Columns: member_id, full_start_date

Code:

```
SELECT member_id,
        DATE_FROM_PARTS (start_year, start_month, start_day)
AS full_start_date
FROM Membership;
```

Table 7: Subscriptions

sub_id	Plan	renewal_date
11	Basic	2025-01-01
12	Premium	2025-03-15

Q7. Extend each renewal_date by 1 year using DATEADD()

Expected colums: sub_id, Plan, extended_renewal_date

code:

```
SELECT sub_id, Plan,
        DATEADD (YEAR, 1, renewal_date) AS extended_renewal
_date
FROM Subscriptions;
```

Output table

sub_id	plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Table 8: Orders

Order_id	order_date
1001	2025-04-15
1002	2025-04-10

Q8. Show current date and difference from order-date

Use CURRENT_DATE and DATEDIFF()

expected columns: order_id, order_date, today_date, date_since order

code:

```
SELECT oder_id, order_date,
        CURRENT_DATE AS today_date;
        DATEDIFF (DAY, order_date, CURRENT_DATE) AS
days_since_order
FROM Orders;
```

Output table

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-17	367
1002	2025-04-10	2026-04-17	372

Table 9: Trainings

training_id	topic	trainning date
1	Safety	2025-01-10
2	Compliance	2025-02-20

Q9. Extract the year from training_date using DATE_PART() or EXTRACT()

Expected columns: training_id, topic, training_year

code:

```
SELECT training_id, topic,
        EXTRACT (YEAR FROM training_date) AS training_
year
FROM Trainings;
```

Output table

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

Table 10: Blog_posts

Post_id	title	Published_on
1	SQL-Tips	2025-04-01 10:15:00
2	Data Cleaning	2025-04-12 16:45:00

Q10. Extract hour and minute from published_on.
Expected columns: post_id, title, hour_published, minute_publishe

Code

```
SELECT post_id, title,
       EXTRACT(HOUR FROM published_on) AS hour_published,
       EXTRACT(minute FROM published_on) AS minute_publishe
FROM Blog_Posts;
```

output table

Post_id	title	hour_published	minute_published
1	SQL - TIPS	10	15
2	Data cleaning	16	45

Table 11: Drivers

driver_id	licence_expiry
301	2025-08-10
302	2023-12-31

Q11. Calculate days left until license expiry using DATEDIFF() and today's date.
Expected columns: driver_id, license_expiry, days_left

Code:

```
SELECT driver_id, license_expiry,
        DATEDIFF (DAY, CURRENT_DATE, license_expiry) AS
days_left
FROM Drivers;
```

Output table:

driver_id	license_expiry	days_left
301	2025-08-10	-250
302	2023-12-31	-837

Table 12: Messages

message_id	sent_timestamp
1	2025-04-19 09:32:45
2	2025-04-18 23:59:59

Q12. Display the current timestamp and calculate seconds since the message was sent.
Expected columns: message_id, sent_timestamp, current_timestamp, seconds_since_sent

Code:

```
SELECT message_id, sent_timestamp,
        CURRENT_TIMESTAMP AS current_timestamp,
        DATEDIFF (SECOND, sent_timestamp, CURRENT_TIMESTAMP) AS
seconds_since_sent
FROM Messages
```

Output table

message_id	sent_timestamp	current_timestamp	seconds_since_s
1	2025-04-19 09:32:45	2026-04-17	~31400000
2	2025-04-18 23:59:59	2026-04-17	~31500000

Table 13: Returns

return_id	return_date
901	2025-04-05
902	2025-04-01

Q13. Add 15 days to return_date use DATEADD() to show restock_date

Expected Columns: return_id, return_date, restock_date

```
SELECT return_id, return_date,
        DATEADD(DAY, 15, return_date) AS restock_date
FROM Returns;
```

Output table;

return_id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 14: Assignments

assign_id	assigned_on
1	2025-03-01
2	2025-03-05

Q14. Convert assigned_on to date using TO_DATE() (If it's

stored as string).
Expected columns: assign-id, assigned_on_date

Code:

```
SELECT assign_id,
        TO_DATE (assigned_on, 'YYYY-MM-DD') AS assigned_on_da
FROM Assignments
```

Output table:

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

Table 15: Meetings

meeting_id ~~SELECT~~	scheduled_time
1	2025-04-19 14:00:00
2	2025-04-19 09:30:00

Q15. Convert scheduled_time to formatted string like 'April 19, 2025 at 2:00 PM' using TO_CHAR().
Expected Columns: meeting_id, formatted_meeting_time.

Code

```
SELECT meeting_id,
        TO_CHAR (scheduled_time, 'Month DD, YYYY "at"
        HH12:MI AM') AS formatted_meeting_time
FROM Meetings;
```

output table

meeting_id	formatted_meeting_time
1	April 19, 2025 at 02:00 PM
2	April 19, 2025 at 09:30 AM